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Conclusions from 3 years of continuous capture plant operation
without exchange of the AMP/PZ-based solvent at Niederaussem –
Insights into solvent degradation management

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Abstract

A many times heard mantra of solvent degradation management in amine-based post combustion capture is “keep the solvent clean” to minimize solvent consumption. It is assumed that the amine losses would decrease by the removal of metals, degradation products, and reactive trace components which are captured from the flue gas, like NO₂ (as potentially driving components of the amine degradation besides dissolved O₂). However, this theoretical hypothesis – based on results from laboratory experiments typically generated with fresh amines – disregards the complexity of the solvent matrix, interaction of potential metal catalysts with degradation products and oxidizing agents, and specific chemical requirements which must be fulfilled before a degradation mechanism can proceed. Degradation of the solvent CESAR1 (aqueous solution of 3.0 molar 2-

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amino-2-methylpropan-1-ol (AMP) and 1.5 molar piperazine (PZ)) is investigated in a unique long-time test campaign (testing time up to now 40 months; 24/7 operation) without replacement of the solvent inventory at the capture pilot plant at the lignite-fired power plant in Niederaussem. Three solvent management strategies with different effect mechanisms are investigated and evaluated: a) removal of only anionic compounds and trace elements (within 75 days solvent inventory treated two times) and anionic as well as cationic compounds and trace elements (114 days, inventory treated four times) from the solvent by ion exchange, b) adsorptive removal of trace elements from the solvent by active carbon in 35% of the operating time, and c) removal of >80% NO₂ by flue gas pretreatment with thiosulfate/sulfite solution (dosing for 2,000 h). The results of the testing program clearly show that “solvent cleanliness” is not a well-defined parameter and that results from laboratory tests, tests without fully representative industrial flue gases, and short-term testing of monoethanolamine cannot be generalized for other solvents and industrial application. These results showcase that specific degradation management considering solvent, capture plant and flue gas quality is reasonable. Overshooting efforts for solvent management are contra-productive and produce unnecessary waste streams, efficiency losses and costs.

Keywords: AMP, piperazine, CESAR1, solvent management, solvent degradation, ion exchange, NO₂ removal, pilot plant, demonstration

1. Introduction

Recently, a guidance was published that proposes available techniques which should be the best for preventing or minimizing emissions and impacts on the environment from post-combustion carbon dioxide capture [1]. According to this guidance, solvent management is a very important factor for the environmental footprint of amine-based CO₂ capture. It is suggested that:

- amine-based solvents should generally be clear and transparent,
- “significant” changes of the solvent color should automatically require the application of reclaiming,
- all non-solvent constituents should be removed from the solvent inventory as fast as they are added during operation to avoid their accumulation.

The underlying idea tries to connect increasing corrosivity of the solvent with increasing concentration of organic acid residues formed by amine oxidation that results in increased release of metal ions from the capture plant components into the solvent which act together with degradation products as catalysts for the amine degradation.

Similar guidelines have also been previously proposed in literature. Derived from experience with the monoethanolamine (MEA) operation at the Technology Center Mongstad, the authors propose color as a degradation indication, suggest “transparent” as a threshold value, and recommend reclaiming in case of any significant color change. Additional degradation indicators and threshold values include: ammonia emission (5 ppmv), heat stable salts (1.5 wt.% of the amine as HSS), metals (5 ppmv), and the sum of degradation products concentration (3 wt.%). Reclaiming is suggested in case any of these variables achieve the threshold limit [2].

The published guidelines also link the formation of nitrosamines in the solvent with solvent management and the concentration of NO_x in the flue gas. It is assumed that solvents using secondary amines would readily form stable nitrosamines in the capture plant in contradiction to primary amines, which would form no such products especially if a high level of reclaiming is used to remove degradation products [1]. Simple approaches for the reduction of the nitrosamine concentration in solvents based on secondary amines, like the thermal decomposition of nitrosamines, are unfortunately not mentioned [3]

However, to a large extent the experimental basis for the underlying assumptions and recommendations is limited to simplified laboratory experiments or testing campaigns at pilot plants which are short compared to the time-scale of solvent degradation processes, and are related to the solvent 30 wt.% aqueous monoethanolamine (MEA) [1,2]. Quite often drastic artificial parameters, regarding partial pressure of gaseous components, temperatures, and concentrations of components in the solvent, are applied for laboratory tests to reduce the testing time and in the hope to achieve strong and conclusive effects (accelerated degradation tests), but the individual effects of the specific parameters must be carefully evaluated for each solvent system [4-6]. Test campaigns at capture plants are often interrupted by exchanges of the solvent inventory caused by operational issues, like corrosion of components or problems regarding the flue gas quality and interfering tasks in testing programs so that the effective testing time on degradation is limited [2]. Buvik et al. have recently assessed 29 individual campaigns on solvent degradation in

18 different pilot plants which are generally not longer than 4,000 h [39]. Also for the slowly degrading CESAR1 solvent the reported testing campaigns have durations of only 1,000 and 1,500 operating hours [40,41].

It is important to validate the proposed solvent management approaches and the hypotheses about solvent degradation against results from real long-time pilot plant or full-scale tests using fully representative, industrial flue gases, over representative periods. This will help to avoid waste streams and efficiency losses by reclaiming that may result from contra-productive overshooting efforts for solvent management. The consequences of the proposed guiding principles are directly connected with the economics of amine-based CO₂ capture, conflicting environmental goals due to cross-media effects (management of solvent consumption, decreased energetic efficiency, generation of waste-streams) and consequences for the permitting, operation and implementation of CO₂ capture plants.

As part of the transatlantic project LAUNCH (Project No. 299662; co-funded by the ERA-NET Accelerating CCS Technologies initiative), a long-time test campaign (testing time up to now 40 months; 24/7 operation) at the CO₂ capture pilot plant at the lignite-fired power plant in Niederaussem with the CESAR1 solvent (aqueous solution of 3.0 molar (26.7 wt.%) 2-amino-2-methylpropan-1-ol (AMP) and 1.5 molar (12.9 wt.%) piperazine (PZ)) is carried out. Three solvent management strategies with different effect mechanisms are investigated and evaluated: a) removal of ionic compounds and elements from the solvent by ion exchange using cationic and anionic exchange resins, b) adsorptive removal of trace elements from the solvent by active carbon, and c) removal of NO₂ from the flue gas by pretreatment with an aqueous thiosulfate/sulfite solution. Also the effect of solvent refills and plant downtime on the degradation rate is evaluated [7, 8].

The results of this unique test campaign of solvent management technologies with CESAR1 are contrary to the above mentioned general recommendations on best available technologies. In this publication it is shown that:

- solvent treatment by active carbon reduced the visible absorbance of the solvent, as well as the iron, nickel, and chloride concentration, but this has no clear effect on the degradation rate and performance of CESAR1,
- anionic compounds, trace components and metal complexes can be effectively removed from the solvent by ion exchange using an anion exchange resin and cations by a cation exchange resin, but removal of anions alone

which explicitly comprises also negatively charged metal complexes of iron, nickel and zinc causes a significant increase of the degradation of CESAR1,

- NO₂ removal from the flue gas upstream the CO₂ absorber by pre-treatment with a thiosulfate solution did not show a reducing effect on the amine degradation, and no significant reduction of the concentration of nitrosamines in CESAR1 could be observed (being a secondary amine, PZ is expected to readily form nitrosamines according to [1]),
- Removal of all contaminants from CESAR1 as fast as they are formed is contra-productive and would cause unnecessary loss of solvent and capture plant efficiency as well as increased volume of waste streams without performance benefits.

2. Methods and Materials

2.1. Capture plant and overview testing campaign with CESAR1 solvent

The post-combustion capture pilot plant at Niederaussem has been operated 24/7 for more than 100,000 hours since 2009. The plant treats 1,550 kg/h of flue gas to capture 7.2 t CO₂/d at the typical conditions given in Table 1. More details can be found in [9, 10].

Table 1. Typical operating parameters and boundary conditions during testing of the solvent management technologies with CESAR1 at Niederaussem (STP - standard temperature and pressure: 273.15 K and 101.325 kPa; dry, 6% O₂ correction).

Parameter	Unit	Value
Flue gas temperature at DCC inlet	°C	62 – 65
Flue gas temperature at absorber inlet	°C	40 – 45
Flue gas flow	m ³ /h, dry (STP)	1,150
CO ₂ content of the flue gas at absorber inlet	vol.%, dry	15.2
O ₂ content of the flue gas after desulfurization	vol.%, dry	5.5
SO ₂ content of the flue gas before desulfurization	mg/m ³ , dry (STP)	1,500 – 3,500
SO ₂ content of the flue gas before DCC	mg/m ³ , dry (STP)	50 – 200
SO ₂ content of flue gas at absorber inlet	mg/m ³ , dry (STP)	<1
Dust content of the flue gas before DCC	mg/m ³ , dry (STP)	<2
NO _x content of flue gas at absorber inlet	mg/m ³ , dry (STP)	100 – 180

NO ₂ content of flue gas at absorber inlet	mg/m ³ , dry	1 – 4
Solvent flow rate	kg/h	2,600 – 3,100
Water circulation in the DCC	kg/h	8,000
pH value of water in the DCC	-	7 – 7.2
CO ₂ -lean flue gas temperature at water wash outlet	°C	40 – 45
Desorber pressure	bar(a)	1.75
Solvent regeneration temperature	°C	120
CO ₂ capture rate	%	90

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112 2.2. Testing campaign with solvent CESAR1

113 The CESAR1 solvent was prepared from the individual amine components in aqueous solution. AMP has been
 114 provided by BCH Brühl - Chemikalien Handel GmbH as 90% solution. PZ was provided as anhydrous chips by
 115 Julius Hoesch GmbH. The piperazine chips were dissolved in pre-heated water and a solution with 30% PZ was
 116 prepared. The final composition of CESAR1 according to the target concentrations of AMP (26.7 wt.%, or 3 M) and
 117 PZ (12.9 wt.%, or 1.5 M) was produced by mixing the two solutions in the solvent tank of the capture plant. No
 118 additives or corrosion inhibitors have been used. Figure 1 shows an overview about the overall testing campaign.

119 The testing campaign with CESAR1 started on 08.04.2019. The capture plant has been operated almost
 120 continuously. During the usually short times when no CO₂ was available from the power plant unit, the capture plant
 121 was operated in “hot mode”. During the “hot mode” the solvent recirculation is maintained and the solvent
 122 temperature in the desorber is kept at approx. 85°-100°C, but the flue gas feed is stopped, hence no CO₂ is captured.
 123 This mode of operation was chosen to further thermally stress the solvent allowing continued solvent degradation
 124 and release of metal ions into the solvent by corrosion. During the phase of a major overhaul of the power plant from
 125 the end of July to the beginning of October 2020, during a short stop in December 2020, and an overhaul in May
 126 2022, the solvent was drained into the solvent tank, which was emptied from fresh solvent before to avoid dilution of
 127 degradation products in the solvent.

128 In the first 1,078 days of operation with CESAR1, the concentration of AMP and PZ and the concentration ratio
 129 AMP/PZ were not kept constant in the solvent, but actions were taken to maintain the water content. For this, the
 130 addition of make-up water to the solvent loop of the capture plant has been adapted according to the results of daily
 131 analyses of the water content. The make-up water is added to the inventory of the water wash section at the top of

the absorber which is connected with the solvent loop. With increasing content of degradation products and trace components in the solvent the amine concentrations decreased, and the solvent circulation was increased (from approx. 2,600 kg/h to 3,100 kg/h within 21,600 h and 3,500 kg/h after 24,500 h) in order to keep the CO₂ capture rate of 90%. After two years of operation and an increase of the solvent circulation of 8% no change of the specific reboiler duty was observed. All in all the sensitivity of the specific reboiler duty regarding degradation and L/G ratio is relatively low for the CESAR1 solvent (specific reboiler duty/solvent flow: day 92: 3,11 GJ/t CO₂/2,741 kg/h; day 171: 3,10 GJ/t CO₂/2,641 kg/h; day 367: 3,10 GJ/t CO₂/2,592 kg/h; day 751: 3,07 GJ/t CO₂/2,842 kg/h). By removal of degradation products in the campaigns on ion exchange from the solvent and the recovery of the initial amine ratio, the solvent circulation rate achieved the initial value. To compensate the solvent losses by sampling, degradation and emissions and to guarantee the operability of the capture plant regarding the necessary solvent inventory, fresh solvent was refilled eight times (between 10 and 25% of the solvent inventory of the capture plant per refill): refills after 79 days (1,896 h), 280 days (6,720 h), 315 days (7,560 h), 458 days (10,992 h), 637 days (15,288 h), 784 days (18,816 h), 970 days (23,280 h) and 1,078 days (25,872 h) from the solvent tank (3 molar AMP, 1,5 molar PZ, aqueous solution; in the last refill the initial amine ratio was adjusted). This means the volume equivalent to one solvent inventory was added to the plant in a period between 600 and 800 days of operation. For comparison, as a result of the ALIGN-CCUS long-term campaign with MEA, the solvent inventory change rate was estimated between 200 and 270 days [11].

A commercially available silicon-based anti-foaming agent (SAG 7133, BASF; injected amount about 50 ml) was added occasionally (one time per month) to the solvent, when fluctuations of the differential pressure in the desorber column indicated incipient foaming. Despite the accumulation of trace components and degradation products, no operational problems by corrosion, precipitation or increasing solvent viscosity have occurred.

Four tests have been carried out to investigate the degradation rate of CESAR1 and the destruction of nitrosamines in the solvent at elevated desorber pressure of 2.4 bar(a) and corresponding reboiler temperature of 130°C (for 7 days after 337 day testing time, for further 5 days after 354 days, 7 days after 822 days, and for 12 days after 912 days during the campaign with NO₂ removal).

After parametric studies using CESAR1 at the beginning of the overall campaign, the proprietary dry bed configuration was almost continuously used from day 400 (9,600 h) in addition to the normal water wash at the top

of the CO₂ absorber column to effectively minimize the AMP and PZ emissions. This significantly reduces the emissions without generating waste streams [8, 12] (reduction of AMP from 35 mg/m³ to < 5 mg/m³ and PZ from 10 mg/m³ to <3 mg/m³). The AMP, PZ and NH₃ concentrations in the CO₂-lean flue gas downstream the water wash section are measured by a GasMET CX/DX 4000 FTIR analyser. Heated lines are used to connect the sampling point and the analyser. The detection limits for AMP and PZ are approx. 1 mg/m³ (STP). The analyser is zeroed with nitrogen once a week. An online THC measurement system (Total Hydrocarbons, Fidamat 6, Siemens) is used to investigate the emissions of organic compounds semi-quantitatively.

The acid wash is a separate process component and no liquids from the acid wash have contact with the water wash or the solvent loop. Therefore, an effect of its operation on the solvent degradation is not possible. During the considered period with tests of solvent management technologies the acid wash was not active.

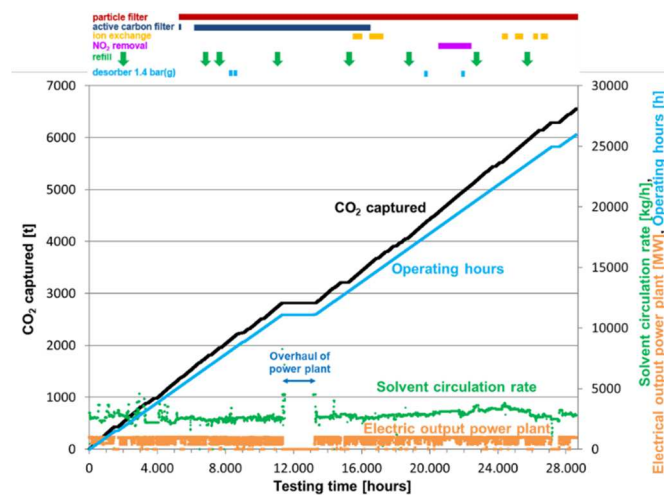


Fig. 1. CESAR1 testing campaign at Niederaussem. The capture plant was operated 28,800 hours from 09.04.2019 and 6,600 t CO₂ have been captured. Applied solvent management technologies: refills by adding fresh solvent (green arrows), phases with elevated desorbent pressure (blue bars, 1.4 bar(g) instead of 0.75 bar(g) applied), and phases with active particle filter (dark red bar), active carbon filter (dark blue bars), solvent treatment using ion exchange (yellow bars), and NO₂ removal (magenta bar) by thiosulfate dosing.

2.3. Test of solvent management technologies

The performance of three solvent management technologies (active carbon, NO₂ removal, ion exchange) and their impact on solvent degradation have been evaluated during the ongoing 40 months testing campaign. Additionally, the effect of solvent refills, elevated desorbent pressure and temperature, and plant downtime on the degradation rate have been assessed. Figure 2 shows the process configuration and integration points of the solvent management technologies in the process of the CO₂ post-combustion capture pilot plant at Niederaussem.

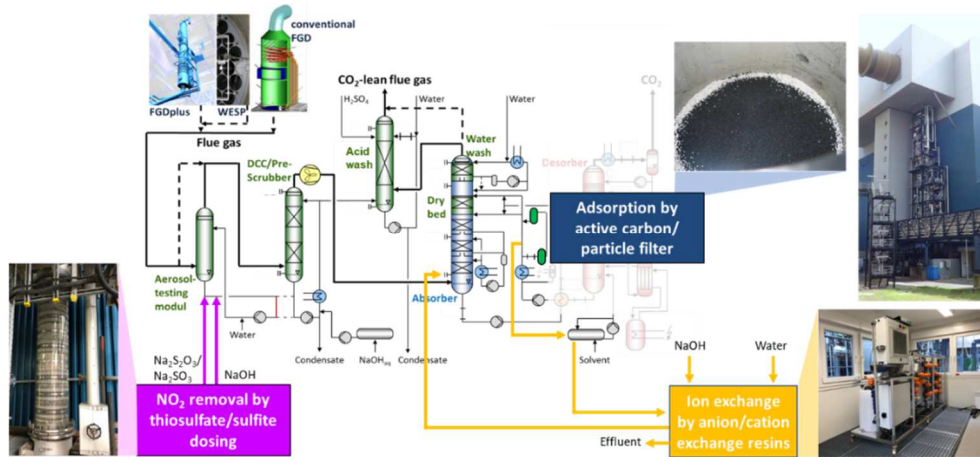


Fig. 2. Integration points of the solvent management technologies in the process of the CO₂ post-combustion capture pilot plant at Niederaussem: NO₂ removal from the flue gas by thiosulfate/sulfite dosing using the aerosol testing module as scrubbing column, active carbon and particle filter in the lean solvent line and feeding of the solvent treated by ion exchange with cation and anion exchange resins into the absorber.

Table 2 shows the total testing times of the individual technologies and the bar chart in Figure 1 the periods of application.

Table 2. Testing periods for the individual solvent management technologies in the 40 months (28,800 h) campaign.

Parameter	Unit	Value
Total testing time with operation of the active carbon filter	h	10,452
Total testing time with NO ₂ removal by flue gas pre-treatment with thiosulfate/sulfite	h	1,192
Total testing time with ion exchange using solely an anion exchange resin	h	1,392
Total testing time with ion exchange using cation and anion ion exchange resins	h	1,536
Total testing time with elevated desorber pressure of 1.4 bar(g) instead of 0,75 bar(g)	h	720

2.3.1. Active carbon filter

The solvent filtration by active carbon uses a commercially available granular material with a particle diameter of 0.6-2,36 mm (Type L 8x30 C from Wocklum Chemie; steam-activated active carbon produced from coconuts by pyrolysis; with a specific BET surface of 1,100 m²/g and a density of 1,050 g/cm³). 10% of the total solvent flow is filtered in the bypass. The active carbon filter is located downstream the solvent/solvent-heat exchange and upstream

the lean solvent cooler, see Figure 2. After 216 days of operation (5,184 h), the active carbon filter was activated for the first time, but only for 12 hours. In parallel, the particle filter was activated and continuously operated after then. After day 252 (6,048 h) of the testing program, the active carbon filter was also continuously operated until day 687 (16,488 h). This means that the active carbon filter was operated only around 37% of the testing time.

2.3.2. NO_2 removal by thiosulfate/sulfite dosing

The NO_x concentration in the feed gas is mainly formed from nitrogen in the lignite and not by thermal formation from the nitrogen content of the combustion air. At lignite fired power plants, the NO_x concentration is controlled by primary measures (air staged combustion) and not by SCR or SNCR as for hard coal-fired power plants. This explains the stronger dependence of the NO_2 formation on the nitrogen content of the lignite and the fact that there is no NH_3 in the flue gas (which would have otherwise resulted from a slip of the De NO_x system). During the NO_2 removal campaign by pre-scrubbing with an aqueous thiosulfate/sulfite solution, the NO_2 concentration in the flue gas was in the range between 1-4 mg/m^3 , dry (STP) up-stream the pre-scrubber. The NO concentration in the feed gas is fluctuating between 100 and 160 mg/m^3 and under full load conditions of the power plant no steplike changes of the NO concentration or extended phases with different concentration levels occur. Only at part load operation (these phases normally last less than 1 day) the oxygen concentration in the flue gas increases from 5.5 to 7.5 vol.%, dry (STP), NO decreases from 140 to 110 mg/m^3 dry (STP) and NO_2 increases from 2.5 to 5 mg/m^3 dry (STP).

In the NO_2 removal campaign at Niederaussem the flue gas pre-treatment comprised the dosing of separate solutions of sodium thiosulfate, NaOH and sodium sulfite in the scrubbing liquid loop of a pre-scrubber (normally used as proprietary aerosol removal module) which was equipped with a high-performance packing (2.24 m high structured packing; specific surface approx. 600 m^2/m^3) to achieve a sufficient high mass transfer of NO_2 from the gas phase into the liquid. This set-up was used instead of dosing into the DCC to avoid the loss of the agents by drained condensate from the DCC. The thiosulfate and NaOH dosing system was integrated into the control system of the capture plant. Alternatively, condensate from the DCC or demineralized water can be used to fill the scrubbing system.

For measuring the NO_2 concentration downstream the DCC at the inlet of the CO_2 absorber, an UV spectrometer was used (SICK GMS 810-DEFOR, measuring range 0.0-50 mg/m^3 , accuracy approx. 0.5 mg/m^3). During the test,

one time per day for a period of 15 to 30 minutes, the pre-scrubber was bypassed to measure the NO₂ concentration without the countermeasure to allow for the evaluation of the effect of the countermeasure. The target concentration of thiosulfate in the scrubbing liquid was 120 mmol/kg at a pH value between 7 – 8 in the scrubbing liquid to achieve an expected NO₂ removal rate of >80%. The thiosulfate concentration was monitored off-line by an analyzer based on Raman spectroscopy and iodometric titration. The results of both methods coincided well. Also, by iodometric titration, the sum of thiosulfate and sulfite concentration in the scrubbing solution was analyzed.

The long-time test of the NO₂ removal by dosing of thiosulfate started at day 854. The analysis of the scrubbing solution resulted in concentrations of thiosulfate up to 15,000 mg/l (134 mmol/l) and sulfite up to 16,500 mg/l (21 mmol/l). By reducing the thiosulfate dosing rate, a minimal sum concentration of thiosulfate and sulfite of 70 mmol/l was identified which is necessary to remove >80% of the NO₂. At lower sum concentrations an increase of the NO₂ concentration in the flue gas at the inlet of the absorber occurred. A decrease of the sulfite concentration in the scrubbing fluid is caused by low SO₂ freights in the flue gas in phases of part load operation of the power plant. A consequence of these phases is a slight increase (typically 0.7 to 1.0 mg/m³) of the NO₂ concentration in the flue gas entering the CO₂ absorber.

2.3.3. Ion exchange with anion exchange resin and serial configuration of cation and anion exchange resins

To evaluate the effects of solvent treatment by ion exchange a mobile system (MIONTEC) was used. Proprietary anion and cation exchange resins provided by the supplier of the ion exchange system were used to remove ionic trace elements from the solvent. In the first campaign the solvent treatment system was first activated after 646 days (15,504 h) of operation. During each ion exchange cycle, approx. 130 l of solvent were treated, corresponding to ca. 4% of the inventory. The lean solvent was taken out of the solvent storage tank which was filled on demand with the needed amount of aged solvent from the solvent loop. After each cycle, the resin was regenerated with 4 wt.% NaOH solution. The cleaned solvent was fed back to the absorber sump. After 17 ion exchange cycles (approx. 2.2 m³ of solvent was treated, or approx. 75% of the total solvent inventory), the solvent treatment was stopped for 18 days to monitor the behavior of the solvent degradation and the concentrations of the trace components in the solvent. After 687 days (16,488 h), the ion exchange module was operated again until in total 48 ion exchange cycles have been carried out on day 721 (17,304 h). Approx. 6.2 m³ of solvent were treated in total or approx. 2 times the

total inventory. It is noted that, after 690 days (16,560 h), the particle and active carbon filter line was closed and emptied into the solvent loop of the capture plant to refill the plant without adding fresh solvent to the solvent hold up. The particle filter was immediately afterwards activated again, but the active carbon filter not until day 1213 (with fresh active carbon of a different quality than the first one). The second campaign was conducted with a serial configuration of two separate columns with cation exchange resin in the first step and an anion exchange resin in the second treatment step. In total 92 ion exchange cycles were realized from day 1,016 (24,384 h) until day 1,131 (27,120 h). The campaign was divided into four periods (sequence: 16 cycles (day 1,016 – 1,030), 24 cycles (day 1,044 – 1,064), 11 cycles (day 1,092 – 1,101), and 41 cycles (day 1,110 – 1,130). In the second campaign, approx. 4 times the solvent inventory was treated. While the anionic exchange resin was regenerated in the same way as in the first campaign, the cation exchange resin needed no regeneration because of the low metal cation concentration in the solvent.

2.4. Analysis of solvent and gas streams

Beside the investigation of the concentrations of trace components and organic acids in the solvent by an independent, certified measurement institute (SGS Chemie-, Industrie- und Spezialanalytik; elements (iron, nickel, sulfur) analysed by ICP-OES; anions of carbonic acids (acetate, free glycolate, formate, oxalate and propionate) and inorganic anions (chlorine, nitrate, sulfate) by ion chromatography (limit of quantification: sulfur <20 mg/kg, iron <1 mg/kg, nickel <1 mg/kg, and other anions <2 mg/kg)) also the concentrations of some amine reaction products were analysed by an independent, certified-measurement institute (HENKEL; ethane-1,2-diamine (EDA), 1-formylpiperazine (FPZ), 2-oxopiperazin (OPZ)), analysed by liquid chromatography and ESI-MS). Additionally, the solvent samples were analysed directly at Niederaussem in terms of water content (Karl-Fischer titration Eco KF Titrator, coulometric titration based on the oxidation of sulfur dioxide with iodine which consumes one molar equivalent of water versus iodine), CO₂-loading (CO₂ titration Eco Titrator Oil from Metrohm), amine content, and qualitatively (uncalibrated), regarding degradation products (Gas Chromatograph (GC) system Agilent 7890B). Also the concentrations of uncalibrated compounds in the chromatogram are semi-quantitatively detected based on the area of the respective signals and an internal standard of the GC. By titration of CO₂-free solvent samples with KOH solution (Titrand from Metrohm; TitriPUR 0.1 mol/l KOH in ethanol, the sample is prepared by boiling out the

CO₂ under a nitrogen atmosphere and with a reflux cooler to avoid any amine or water losses; the water content is examined by Karl-Fischer titration) the acid number was determined which represents the total concentration of acids (inorganic and organic) and is quoted as formic acid equivalents (consumption of 1 mol KOH at the equivalence point corresponds to 1 mol (46.03 g) formic acid equivalents). Acid number and the concentration of uncalibrated compounds are integral values for the degradation products in the solvent and allow to evaluate the level of degradation, even if not all degradation products are known. Figure 3 shows that the acid number and the content of uncalibrated compounds are highly correlated which indicates that both semi-quantitative values represent the level of solvent contamination well and that no important “hidden compounds” exist which are dominating the degradation processes in the unknown background of the solvent matrix.

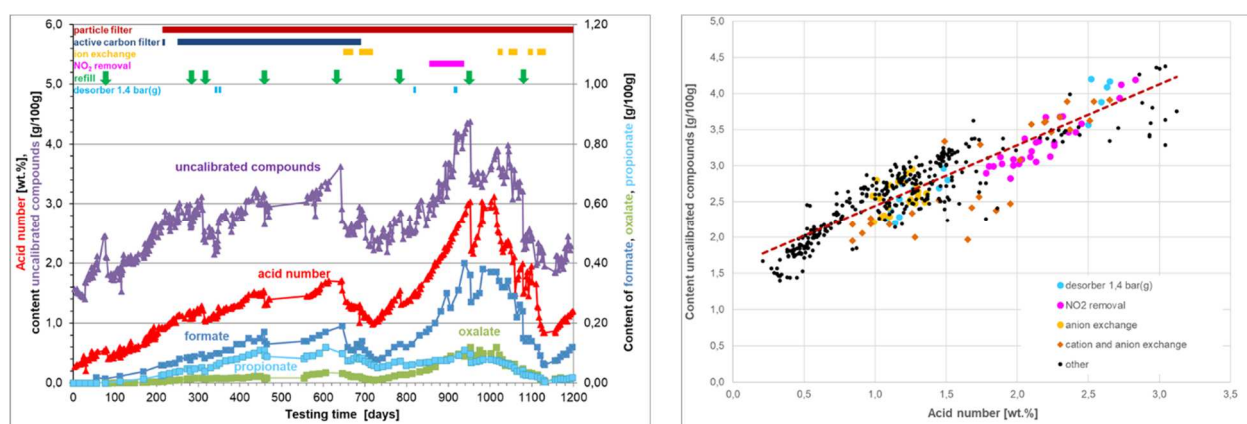


Fig. 3. Acid number and content of uncalibrated compounds are semi-quantitative and integral measures for the level of contamination of the solvent (left diagram). They comprise also unknown species in the solvent. Both are highly correlated (right diagram). Taking into consideration the effects of solvent management technologies on both measures the scattering is relative low.

As expected, at lowest acid numbers (which means at the beginning of the campaign with fresh CESAR1) the content of uncalibrated compounds is lower than for the aged solvent. In addition, ion exchange removes slightly more acids from the solvent than uncalibrated compounds taking the relative changes into consideration.

The analyses regarding the nitrosamine concentrations in the solvent have been carried out by HENKEL using in-house analytical methods. The analysis error is estimated at +/-20%. The matrix of the aged solvent means a challenge for the accuracy of the analysis results. For the analysis of the concentration of N-nitrosodiethanolamine (NDELA) the sample preparation comprised the derivation with N-methyl-N-(trimethylsilyl)trifluoroacetamide. The measurement is based on gas chromatography (GC) and a chemiluminescence detector after addition of an internal standard (N-nitrosodiisopropanolamine). Results of the double measurement were averaged. Also

mononitrosopiperazine (MNPZ) was analysed with GC/chemiluminescence (the response factor was stoichiometrically derived and evaluated against an internal standard). The concentration of total N-nitrosamines comprised the dissociation of the NO group of all nitrosamines in the solvent and detection by chemiluminescence. To compensate for matrix interferences the standard addition method was applied, using NDELA for spiking. As the individual nitrosamines in the solvent and their molecular weights are not known the molecular weight of the N–N=O-group (44 g/mol) as structural element of all nitrosamines is used for the calculation of the mass-related content of total N-nitrosamines. Also here the results of the double measurement were averaged.

3. Results

3.1. Stability of CESAR1 compared with aqueous 30wt.% MEA

Since very often the analysis of solvent degradation rate, degradation mechanisms and the effect of solvent management technologies are evaluated only related to 30 wt.% aqueous MEA as solvent, it is important to underline the huge differences between MEA and CESAR1 (and other solvents). Figure 4 contrasts the accumulation of degradation products, trace components and applied solvent management technologies for the ongoing CESAR1 campaign and a long-time test with 30 wt.% MEA solvent at the capture plant at Niederaussem [13, 14]. Note that the duration of 18 months of testing with MEA is different than that of the campaign with CESAR1, as shown in Figure 4.

Based on the accumulation of organic acids in the solvent as the measure for degradation, CESAR1 can be categorized as a solvent with linear degradation behavior while the degradation of MEA proceeds strongly non-linear. After 400 days (9,600 hours) of testing, the specific solvent consumption was approx. 0.45 kg/t CO₂ for CESAR1. This value takes into account the amine concentration of the solvent, the solvent filling levels in different components of the capture plant, the amount of solvent removed from the process for analysis by sampling (approx. 0.05 kg/t CO₂) and the batch-wise addition of CESAR1 into the process from the storage tank (refills). Therefore, the calculated amine consumption comprises the sum of losses by emissions, degradation and sampling.

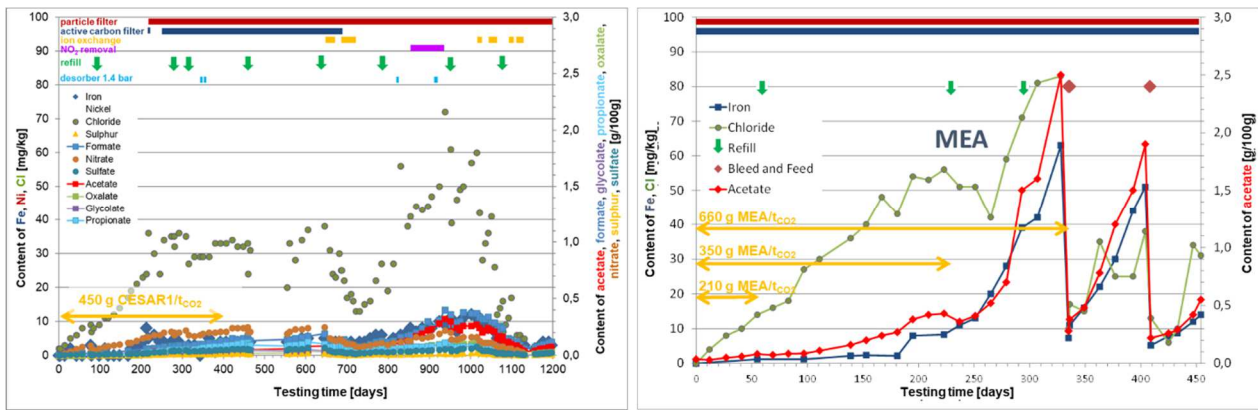


Fig. 4. Comparison of the accumulation of degradation products and trace components in the solvent in two long-time testing campaigns with CESAR1 (left) and aqueous 30wt.% MEA solvent (right).

Although the loss of MEA at the Niederaussem post-combustion capture pilot plant progressed very slowly in comparison to testing campaigns at other pilot plants, after 328 days of operation the MEA consumption was approximately 0.66 kg/t CO₂ [14]. After that time, the targeted capture rate of 90% could not be achieved any longer by increasing the solvent flow rate.

Between 200 and 270 days of operation in the campaign with MEA, self-acceleration of the degradation occurred and an exponential degradation regime was observed. A complete exchange of the solvent inventory (which was not realized) would be connected with additional 0.38 kg/t CO₂ of MEA after 328 days or 0.46 kg/t CO₂ after only 270 days (see Fig. 4).

MEA shows a higher corrosivity than CESAR1 according to the higher iron concentration in the solvent. After 328 days of operation and using the active carbon filter continuously, 63 mg/kg iron accumulated in the MEA solvent. For the CESAR1 solvent, the iron concentration after 329 days was only 3 mg/kg and the highest concentration in the overall campaign was 12 mg/kg before ion exchange was applied. For MEA, the solvent management measure “Bleed and Feed” was applied two times (in solvent purge, a portion of contaminated solvent is removed and replaced by fresh solvent) to reduce the concentration of contaminants. Taking the iron concentration as a tracer for the concentration of solvent contaminants, their content was reduced by 88% and 92%, but without a significant reduction of the degradation rate [9]. Due to the loss of useful solvent and the missing effect on the amine

degradation rate despite the very high dilution of contaminants, “Bleed and Feed” was discarded as a solvent management measure for CESAR1.

The proposed guiding principle “all non-solvent constituents should be removed from the solvent inventory as fast as they are added during operation to avoid their accumulation” would require to maintain very low concentrations of contaminants. To achieve the same concentrations by Bleed and Feed would require extreme dilution factors which might be unrealistic in industrial capture plant operation and not necessary at all for more stable amines than MEA [1, 2]. According to this work’s findings, an unequivocal concentration limit that must be adhered to keep the solvent “clean” cannot be specified neither for the individual nor for the sum of contaminants, while no evidence exists from long-term tests with real flue gas that such a limit exists.

In the following sections the effect of the individual solvent management technologies is presented and discussed using graphs that show the increase of the acid number (difference of the formation and the degradation of acidic compounds in the solvent) as measure for the solvent degradation behavior. The linear degradation behavior of CESAR1 does not mean that the gradient of the increase of the acid numbers remains constant after the application of the solvent management technology. Therefore, linear regression has been used to quantify the change of the increase rate of the acid number which is expressed by the gradient of the line. The longest possible evaluable testing periods have been used for each technology. Figure 5 shows that the analysis of shorter periods +/-10 to 20 days (“local gradients”) for the linear regression does not change the results and trends.

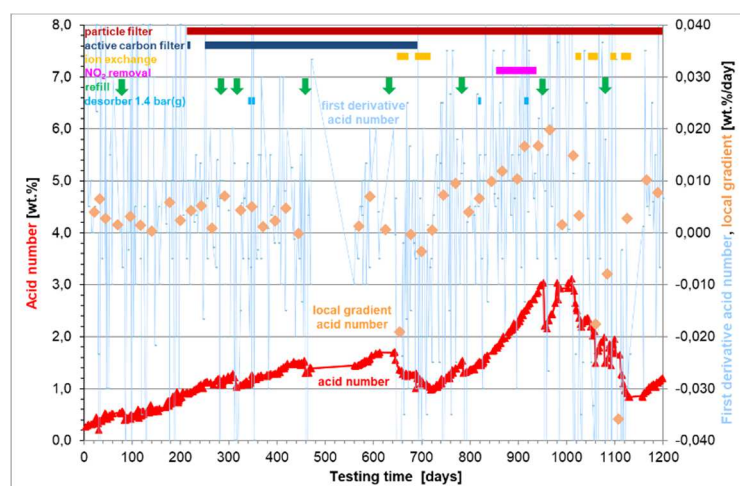


Fig. 5. Acid number, its first derivative and “local gradients” (gradients received by linear regression for short periods +/-10 to 20 days around the individual point) can be used to describe changes in the increases rate of acidic degradation products.

3.2. Effect of active carbon

Solvent treatment by active carbon is usually applied in capture plants to avoid operational issues - like foaming - although the exact effect mechanism is not completely clear. Active carbon effectively adsorbs specific contaminants because of its large surface area, large number of surface-active adsorption sites (also functional groups like O–H and/or N–H groups are available as well as metal oxides and hydroxides), and high adsorption capacity. As the proposed guiding principle “amine-based solvents should generally be clear and transparent” links the color and transparency of the solvent with the degradation rate, application of active carbon leading to color removal should also reduce the amine degradation rate and the accumulation of the degradation products in the solvent [1]. However, for the hypothesis that coloring contaminants are accelerating the degradation and lowering the performance of the solvent, no evidence is resulting from the long-time test of CESAR1 at all. When the active carbon filter is operated, it removes coloring contaminants from the solvent, see Figure 6. Additionally, the concentrations of iron and nickel are reduced (iron from 8 to 4 mg/kg and below the detection limit after two refills; nickel from 5 to 3 mg/kg and after the refills down to 2 mg/kg). Also the accumulation of chloride in the solvent is stopped which is exactly what is proposed for these three specific contaminants named in [1] to ensure satisfactory capture plant operation, see Figure 7.

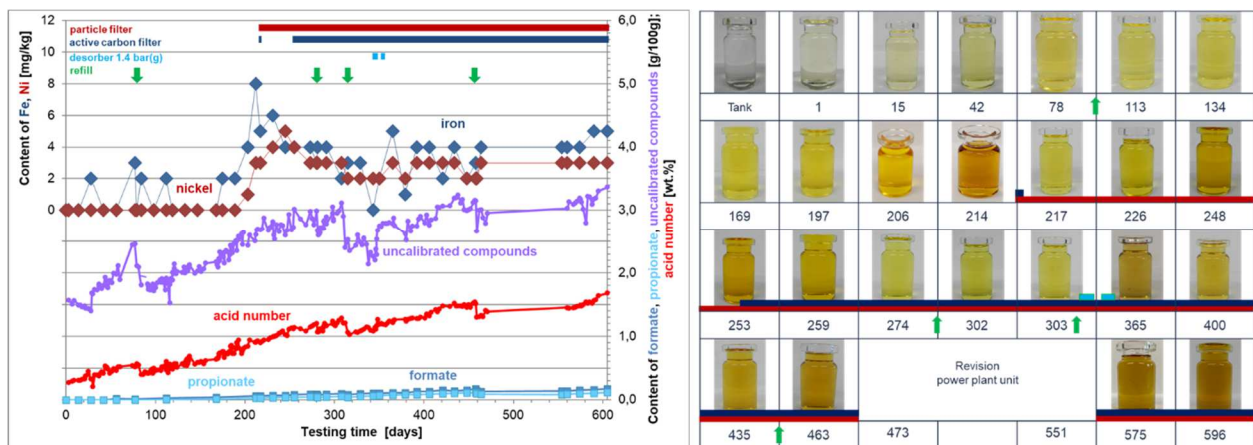


Fig. 6. Active carbon is able to remove a part of the solved iron and nickel from the solvent (left diagram). Although the coloring components are not known, it is obvious that application of active carbon brightens again the solvent (right diagram). The period of operation of the active carbon filter is marked in both diagrams by the dark blue bar.

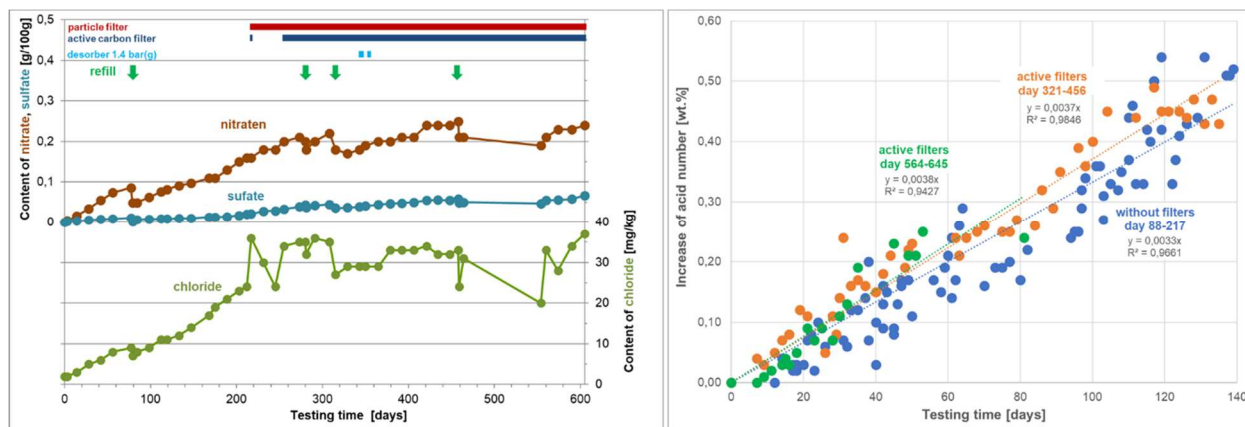


Fig. 7. Active carbon is able to remove a part of chloride from the solvent, but no nitrate and sulphate anions (left diagram). There is no effect of the active carbon filter on the rate of accumulation of degradation products in the solvent – expressed by the increases of the acid number – (right diagram) before and after the activation of the filter.

In contradiction to the proposed guiding principle, there is no effect of active carbon on the increase of the concentration of degradation products (the increase represents the difference between formation and decomposition of the respective degradation product). From the results it even cannot be deduced that the removal of iron and nickel complexes are responsible for the color change of the solvent as both metal cations and some cations from other transition metals have overlapping absorption wavelengths. Additionally, the absorption wavelengths is depending on the complexation of the metal cation. The results indicate that the color and transparency of the solvent is meaningless for evaluating the solvent performance and the solvent degradation rate. For more than 600 days (and before the concentrations are changed by application of ion exchange), the accumulation of amine degradation products in the solvent follows a linear trend. Figure 6 shows especially that there is no correlation between the acid number (indicator for degradation) and the metal content in the solvent, indicating that the role of metal ions in the solvent degradation must be reassessed.

3.3. Effect of anionic ion exchange

The results of the removal of anions from the solvent by an anion exchange resin are also contrary to the “keep the solvent clean” operational philosophy. The concentration of anions could be significantly reduced by more than 50%, see Table 3 and Figure 8. However, directly after the completion of the anion exchange campaign, the rate of increase of the acid number doubled and the rate of increase for uncalibrated compounds even rose by a factor of

4.6. While the increase of the acid number remained high, in the following 100 days of operation, the accumulation of uncalibrated compounds slowed down, but remained higher than before the anion exchange, see Figure 9.

Table 3. Reduction of the concentrations of contaminants in the CESAR1 solvent by the application of anion exchange.

Concentrations	Fe [mg/kg]	Ni [mg/kg]	chloride [mg/kg]	nitrate [wt.%]	sulfate [wt.%]	formate [wt.%]	acetate [wt.%]	glycolate [wt.%]	oxalate [wt.%]	propionate [wt.%]
Before anion exchange	5	3	38	0.25	0.066	0.19	0.083	0.038	0.033	0.099
After anion exchange	3	2	13	0.079	0.014	0.079	0.038	0.018	0.01	0.06
Reduction [%]	40	33	66	68	79	58	54	53	70	39

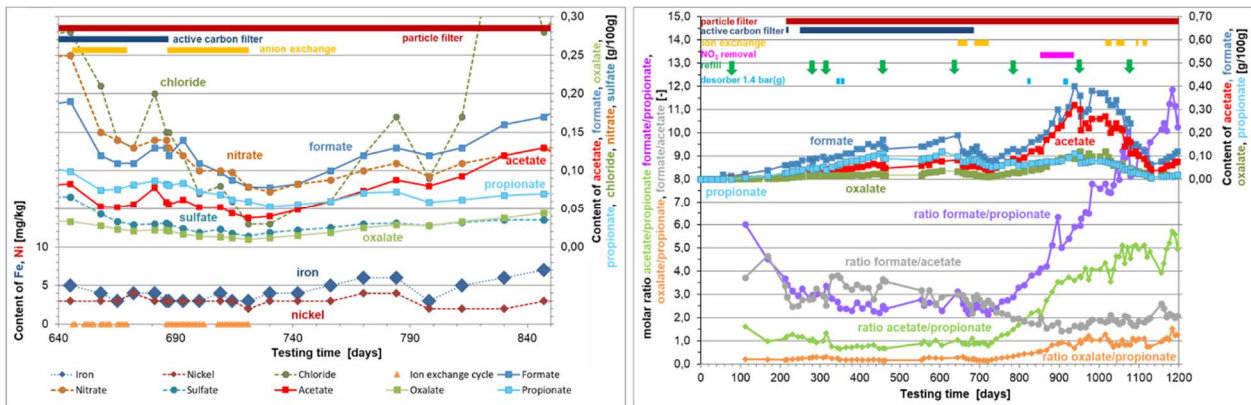


Fig. 8. Left diagram: Removal of trace components and oxidative degradation products from the CESAR1 solvent by ion exchange using an anion exchange resin and fast increase of their concentrations after the solvent treatment. In total 48 cycles were applied (start day 645 (15,480 h), end 721 (17,304 h). Before the start of the last 31 ion exchange cycles, the active carbon filter was deactivated. Right diagram: Reducing the content of anionic contaminants like formate, acetate, oxalate and propionate caused a steep increase of the solvent degradation and changes in the molar ratios of oxidation products in the solvent.

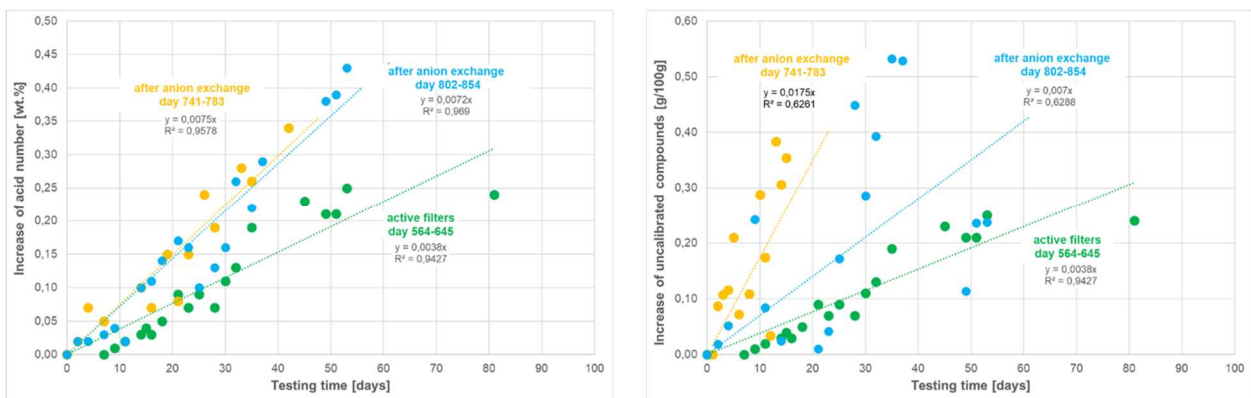


Fig. 9. Although significant amounts of complexed metal ions and anionic inorganic and organic anions have been removed by the anion exchange resin the increase of the acid number (left diagram) and the uncalibrated compounds (right diagram) was much faster than before the anion exchange started (green line: “active filters” is the period before the anion exchange).

The anionic ion exchange resin can remove a significant portion of the iron content from the solvent, but less nickel. Also the concentration of zinc and chromium can be significantly reduced [10]. This result implies that metal cations which are dissolved in the solvent partially exist as anionic complexes and underpins also the importance of the solvent matrix for the properties and the catalytic effects of metal ions on amine degradation, as the properties of the ligands can control the redox behavior and the catalytic activity of a complexed metal ion.

The purpose of the proposed operational philosophy “keep the solvent clean” would be to keep solvent degradation controlled at a low rate. The steep increase of the level of degradation observed directly after the anion exchange campaigns in this work is the exact opposite effect. The increased degradation rate occurred although the concentration of anionic compounds and iron have been reduced to levels which are lower than 12,000 operating hours before. Figure 8 shows that the accumulation of propionate in the solvent differs from that of acetate and formate (as a reminder, the measured content is the result of the simultaneous formation and degradation of propionate). It remains unclear if the formation of propionate was slowed down or if the degradation product propionate itself is faster degraded. Propionate might be formed as follow-up product from the degradation products acetaldehyde and formaldehyde [15]. Also the ratio formate/acetate shows a significant change after applying anion exchange. In any case, this is an indication that cleaning the solvent by removing anionic compounds not only changed the total rate of solvent degradation, but also the relative rate of different degradation pathways.

However, this failure of the concept to remove contaminants from the solvent to reduce solvent losses, together with the confirmation that metal ions which are potentially catalyzing degradation reactions exist as negative charged complexes in the solvent, opens the way to better understand and control the degradation mechanisms in amine-based solvents.

Also bidentate chelating ligands which are able to form strong complexes are existing in aged CESAR1, like ethylenediamine (EDA) (up to 0,46 wt.% or 77 mmol/kg in the test campaign) or oxalate (up to 0,12 wt.% or 14 mmol/kg), in sufficient amounts for the complexation of iron (maximum concentration 12 mg/kg or 0,21 mmol/kg)

and other metal cations in the solvent, see Figure 10. The EDA concentration reached a maximum concentration after ca. 430 days of operation and decreased linearly in the following campaign. The concentration of oxalate increased very fast after the anion exchange campaign, by factor of 100.

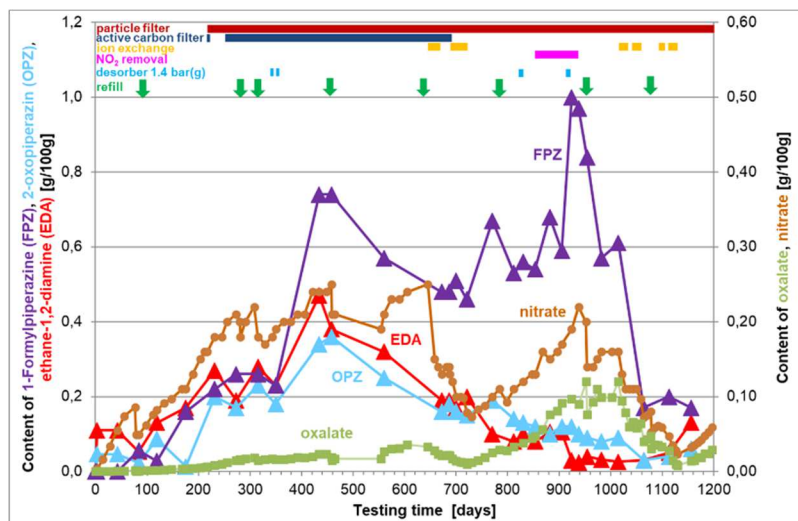


Fig. 10: Accumulation of the degradation products EDA: ethylene-1,2-diamine; FPZ: 1-Formylpiperazine; OPZ: 2-Oxopiperazine, nitrate and oxalate in CESAR1 during 1200 days of testing. Phases with elevated desorb pressure (blue bars, 1.4 bar(g) instead of 0.75 bar(g)) had no effect on the concentrations of EDA or oxalate, but on that of FPZ.

3.4. Effect of serial ion exchange with cation and anion exchange resin

The removal of contaminants by applying a cation and an anion exchange resin in serial configuration reduced the concentrations of impurities and metals to levels which have been determined after 130 and 210 days of operation at the beginning of the campaign and before the first time particle and active carbon filters have been activated, see Figure 11. The achieved reductions are between 73 (iron) and 100% (nickel), see Table 4.

The removal of contaminants by serial application of cation and anion exchange resins resulted in a reduction of solvent degradation, see Figure 12. The increase rate of the acid number has been reduced to 1/3 of the rate before ion exchange. About 70 g of amine per ion exchange cycle have been lost which is only 0.0058% of the amine inventory. Also the concentration of FPZ is significantly reduced in the phase of cation and anion exchange.

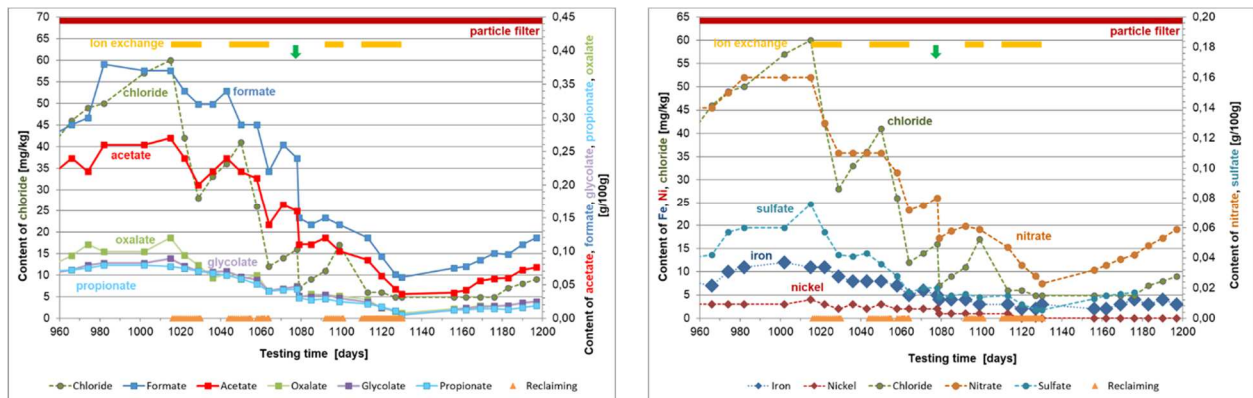


Fig.11: Removal of oxidative degradation products (left diagram) and metal ions and inorganic anions (right diagram) from the CESAR1 solvent by 92 ion exchange cycles (orange triangles) using a serial configuration of cation and anion exchange resins. The concentrations of metal ions, inorganic anions and organic acid residues have been reduced to levels below the concentrations determined in the period <210 days of operation at the beginning of the overall campaign.

Table 4. Reduction of the concentrations of contaminants in the CESAR1 solvent by the application of anion and cation exchange.

Concentrations	Fe [mg/kg]	Ni [mg/kg]	chloride [mg/kg]	nitrate [wt.%]	sulfate [wt.%]	formate [wt.%]	acetate [wt.%]	glycolate [wt.%]	oxalate [wt.%]	propionate [wt.%]
Before cation/anion exchange	11	4	60	0.16	0.076	0.37	0.27	0.089	0.12	0.077
After cation/anion exchange	3	0	<5	0.023	0.0054	0.061	0.036	0.005	0.0076	0.005
Reduction [%]	73	100	92	86	93	84	87	94	94	94

It is instructive to compare the rates of accumulation of the acidic components in fresh CESAR1 (from campaign day 0 to 74), in the phase after the first refill and before the particle and active carbon filters were activated for the first time (from campaign day 77 to 216), and in the phases after the application of anion exchange (from campaign day 741 to 783 and after refill at day 784 from 802 to 854), and cation and anion exchange (from day 1,159 to 1,200). After removing anions and cations from the solvent, the degradation rate, expressed by the increase rate of the acid number, is similar to the rate after anion exchange only. The lowest increase rate is not found for the fresh, most clean solvent, but for mildly aged solvent without any application of solvent management. Taking into consideration that before the stepwise increase of the acid number after the anion exchange (day >741), the increase rate of the acid number was lower than at the beginning of the overall campaign (day 0 to 74), it becomes clear, that preventive cleaning of the CESAR1 solvent does not lead to an improvement.

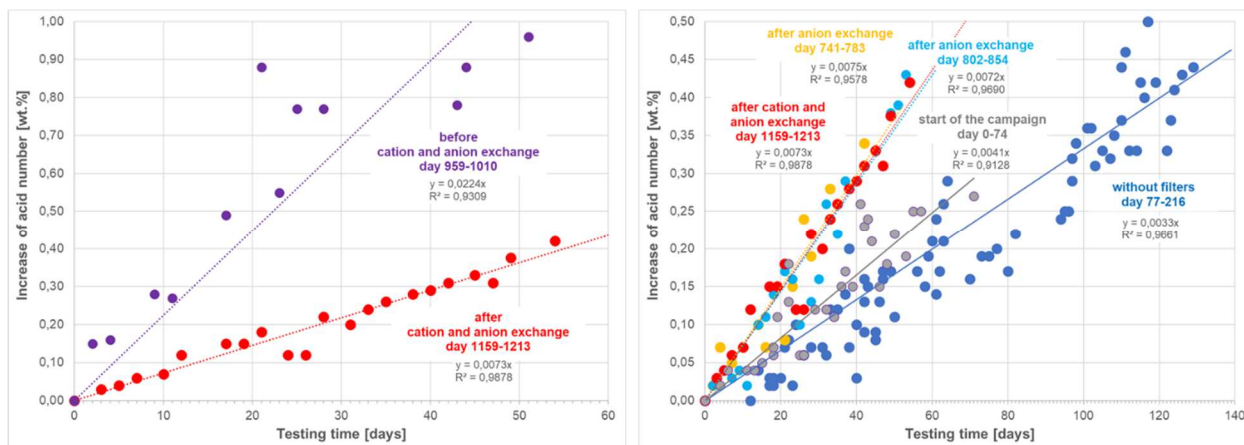


Fig. 12. Application of ion exchange with serial configuration of cation and anion exchange resins significantly reduced the formation and accumulation of acidic degradation compounds in CESAR1 (left diagram), but the achieved increase rate of the acid number shows no difference compared to the rate after anion exchange. The lowest rate of increase is not found for fresh solvent, but for an mildly aged solvent.

3.5. Effect of NO_2 removal by flue gas treatment with thiosulfate

The NO_2 removal by >80% had no effect on the level of ammonia emissions in the CO_2 -lean flue gas downstream the water wash of the CO_2 absorber, see Figure 13. This is an indication that the impact of the thiosulfate/sulfite dosing on oxidative degradation was negligible. The temperature in the pre-scrubber was ca. 63°C. Condensate formation in the insulated pre-scrubber was negligible.

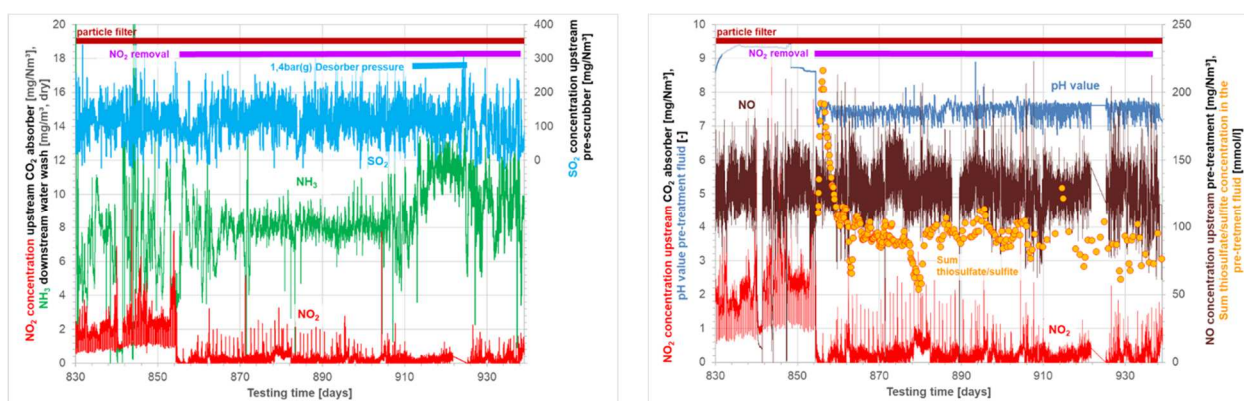


Fig. 13: NH_3 emission from CO_2 -lean flue gas downstream the water wash section during the test of the NO_2 removal by thiosulfate/sulfite dosing (left diagram) into the pre-treatment column and effect of the SO_2 concentration in the raw flue gas on NO_2 capture. The NO concentration is between 100 and 150 mg/Nm^3 (right diagram).

The development of the concentration and the accumulation of degradation products and trace compounds during the phase with thiosulfate dosing shows that the formation rate of the oxidative degradation products formate, acetate, and oxalate was not reduced and also the trends of the concentrations of uncalibrated compounds and the acid number show no decreased solvent degradation, see Figure 14.

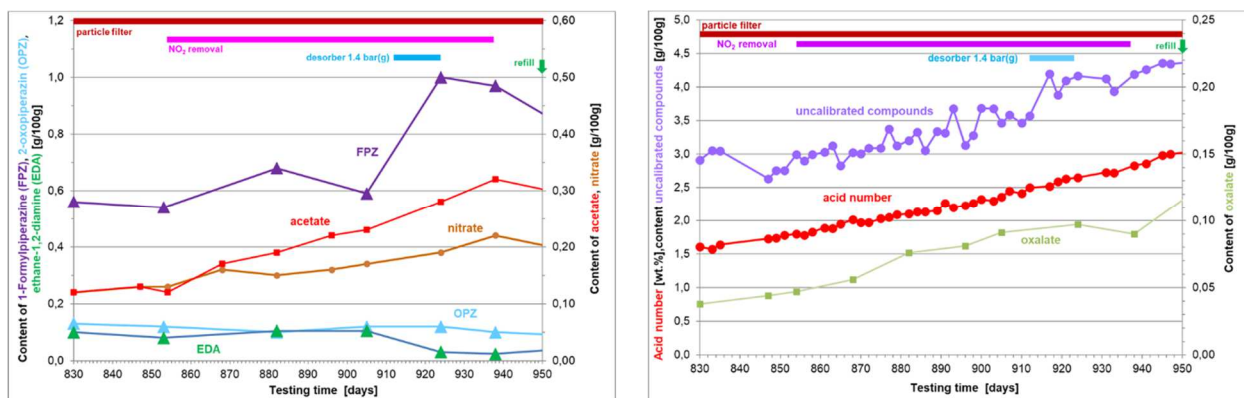


Fig. 14: Accumulation of EDA, FPZ, OPZ, acetate and nitrate during the campaign on the effect of NO₂ removal (left diagram). The strong increase of the FPZ concentration is caused by the elevated desorber temperature. There is no effect of the NO₂ removal on the time course of the acid number and the accumulation rate of uncalibrated compounds (right diagram).

ONONO₂ is believed to react with amines to form N-nitrosamines, whereas the reaction of O₂NNO₂ with amines should form N-nitramines [16]. According to this hypothesis, which focus on the gaseous species and not on the reactive compounds that are dissolved in the solvent forming the nitrosamines, a reduction of the (NO₂)₂ concentration in the flue gas should result in a decrease of the formation rate of nitrosamines in the solvent. Taking into consideration that also nitrosamines are degrading in the solvent by oxidation, radical attack or thermal decomposition, a stable or decreasing concentration of nitrosamines is expected in the 2,000 hour-period of flue gas pre-treatment with thiosulfate solution. The nitrosamine concentrations (total nitrosamines and MNPZ) in the solvent showed no decreasing trend during the campaign, keeping in mind that the analysis error for nitrosamines is estimated at +/-20%. However, using the well-proven degrading effect on nitrosamines of an increased desorber temperature, see Figure 15, a significant reduction of the MNPZ concentration could be achieved and most likely a slight reduction of the total nitrosamine concentration. Additionally, this moderate thermal treatment caused no accelerated accumulation of acidic compounds or uncalibrated compounds. Only the formation of FPZ was accelerated by this effective solvent management measure against nitrosamines. In contrast to MEA-based solvents, the concentration of the nitrosamine NDELA is comparatively low or not detectable for CESAR1.

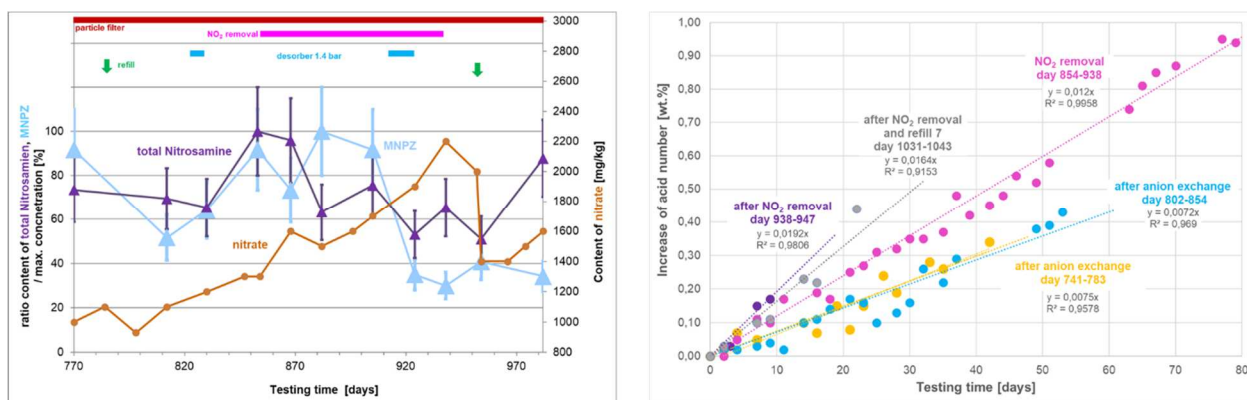


Fig. 15: Accumulation of nitrosamines and nitrate during the test of NO₂ removal by thiosulfate dosing (left diagram). Taking into consideration the analysis error of $\pm 20\%$ no significant changes of the relative concentrations of MNPZ and of the total nitrosamines (actual concentration normalized to the maximum concentration) could be observed. The accumulation of acidic compounds in the solvent, expressed by the acid number, was not reduced, but increased compared to the period after anion exchange (right diagram).

Overall, the increase of the concentration of the acid number was slightly accelerated during the NO₂ removal campaign compared with the phase after the anion exchange. After the NO₂ removal campaign, the increase of the concentration of acidic contaminants was slightly faster compared with the phase of NO₂ removal, but followed also a linear trend.

4. Discussion

4.1. Role of metal ions for amine oxidation

At first glance, the process of the oxidation of amines, but also of their degradation products which are accumulating in the CO₂ capture solvent, seems to be a simple reaction between the residual oxygen in the flue gas and the organic compounds. After reaching the thermodynamic equilibrium, it can be expected that in an alkaline solution dissolved HCO₃⁻, NO₃⁻ and H₂O are received as oxidation products, because of their strong negative Gibbs functions for formation of the compounds ($\Delta_f G^\circ_{molar}$ at 298.15K: H₂O_(liquid) -237.1kJ mol⁻¹; HCO₃⁻_(aq) -586.8kJ mol⁻¹; NO₃⁻_(aq) -108.7kJ mol⁻¹[17]). However, these oxidation reactions do not proceed spontaneously and it is a matter of fact that the oxidation mechanism is highly complicated and comprises several elemental steps [18]:

- dissolution of the gaseous oxidizing species in the flue gas (e.g. O₂, NO, NO₂/N₂O₄, SO₃) into the solvent in the CO₂ absorber comprises diffusion of the oxidant from the gas phase to the flue gas-solvent interface,

- transfer of the species across the interface hydrolysis/ionization of the reactant in the solvent phase if possible,
- diffusion of the ionic and non-ionic species inside the solvent, and a wide range of possible chemical reactions inside the solvent.

An explanation why organic compounds in the solvent are not instantaneously oxidized by O_2 is the fact that organic substrates (amines, alcohols, ketones and acids) in the ground state all have closed shell electronic structures, and therefore are in a singlet state, while O_2 in the flue gas is a diradical (triplet O_2 , ground state with the two unpaired electrons is the triplet spin state). Chemical interactions are governed by various selection rules, among which the ones due to Wigner formulate the constraint of overall spin-conservation [19]. For the reaction to occur, i.e. to preserve the spin, the transition state has to be a combination of the spins of the reactants and the products. That is the algebraic sum of the spins S of the reactants has to have at least one term in common with the algebraic sum of the spins of the products ($|S_{\text{Reactant}} \pm S_{\text{Oxygen}}| = |S_{\text{Product1}} \pm S_{\text{Product2-Oxygen}}|$) [20, 21]. An implication of these spin-selection rules is the inertness of 3O_2 , if the other reactant and the product are spin singlets, as this reaction is spin-forbidden.

Reactions between organic substrates in singlet state with oxygen would be allowed by spin selection rules when the oxygen molecule is in an excited singlet state with paired electrons in different π^* orbitals. The necessary excitation energies are $156.9 \text{ kJ mol}^{-1}$ for $O_2(^1\Sigma_g^+)$ and $94.14 \text{ kJ mol}^{-1}$ for $O_2(^1\Delta_g)$ related to the ground state $O_2(^3\Sigma_g^-)$. Singlet O_2 can be photochemically produced by electronic energy transfer from photo-excited sensitizers or by thermolysis of peroxides. However, due to the lack of radiation with appropriate wavelength and dye-sensitizers, singlet oxygen seems to be irrelevant for the solvent degradation.

There are different options to circumvent the reaction barrier for triplet oxygen [22]. Metal complexes can provide spin-allowed pathways for reaction, due to their abilities to act as radicals with varied spin multiplicities if the number of electrons on a ternary complex (metal ion + amine + O_2) remains constant throughout the reaction [23]. Here, the reaction of 3O_2 with a paramagnetic metal $LM^{(n)+}$ ion (L represents complexation of metals like $Fe^{2+,3+}$, Cu^{2+} , $Co^{2+,3+}$, $Ni^{1+,2+,3+}$ by one or several ligands) can follow an activation route by sequential electron transfer summarized as: triplet oxygen 3O_2 , superoxide O_2^- , peroxide O_2^{2-} and oxide O^{2-} . These reduction intermediates are weak acids and may exist protonated in the rich amine solution. Alternatively, free-radical

formation or autooxidation can be catalyzed by metal ions. Here, a metal ion catalyzes the formation of free radicals, and thus the initiation of chain reactions that are responsible for the oxidative amine degradation. Moreover, the formation of hydroxyl radicals which react with the amine by hydrogen abstraction and formed by protonation of peroxides is catalyzed. Finally, the formation of organoperoxy radicals due to the direct one-electron oxidation by a metal ion can also be catalyzed [24].

After 25,000 testing hours at the capture pilot plant at Niederaussem, the concentration of iron in the CESAR1 solvent reached concentrations around 10 mg/l (0.19 mmol/l). Assuming that all iron atoms are present as dissolved Fe^{3+} ions which oxidatively attack AMP and PZ according to their molar ratio, the calculation of the specific amine loss results in only 0.016 g/l (0.011 g/l AMP and 0.005 g/l PZ). As the amine concentration is 400 g/l, it becomes immediately clear that an oxidation mechanism based on iron is negligible in the case that Fe(III) is only consumed (and transformed to Fe(II)) and not somehow recovered. It is very important to note that the reduced metal must be regenerated by the oxidation agent to act as a catalyst. The regeneration can be performed by oxygen from the flue gas which is dissolved in the CESAR1 solvent under formation of Fe(III). Therefore, the iron cation and other metal ions could act as a catalyst in the oxidation processes in a spin-allowed reaction. However, metal cations do not exist as naked ions in the solvent, but are complexed with water, amines and degradation products. If this would not be the case, precipitation of ferric hydroxide $\text{Fe}(\text{OH})_3$ would occur at the high pH value of the amine solvent [10]. The donation of electron density from the ligand to the metal centre is an important factor for the properties, stability, and reactivity of the complex as well as for the lability of the remaining ligands in a mixed ligand complex. In chelate complexes polydentate ligands, like polyamines (ethylenediamine), aminocarboxylates (e.g. amino acids, like glycine), and polycarboxylates (e.g. oxalate), additionally show the effect of ligand steric hindrance on the mechanism or the kinetics as the coordination cage surrounding the metal cation causes steric constraints around the metal centre.

It is therefore postulated that different amines – and degradation products – may deactivate metals which could otherwise act as catalysts for amine degradation. Hence, and based on the evidence from this work, it seems premature to prescribe a generalized threshold for metal concentration in amine solvents.

638 4.2. The role of ligands on catalytic effect of metal ions

639 The reactivity of metal ions in chemical and biological systems comes from the fact that the displacement of
640 ligands is an imported step in complex-formation reactions and in many redox processes. Intensive investigations of
641 the reactivity of Fe(III)/Fe(II) complexes in amine solutions have been carried out in connection with treatment of
642 sour gas streams to remove H₂S by oxidation with O₂ or to reversibly bind NO in gas cleaning applications. Here,
643 especially iron chelate complexes with ligands from the group of aminocarboxylates, like the prominent hexadent
644 ethylenediaminetetraacetate (edta), but also small amino acids have been studied as absorbent [23, 25-27].
645 Alkanolamines are not used in these industrial processes because of their lower oxidation stability, but this does not
646 mean that they could not form chelate complexes. The chelated iron is the oxidation catalyst in aqueous solution and
647 the ligands determine the rate and overall equilibrium constants, as well as the nature of the substitution mechanism
648 by which O₂ or NO coordinate to the selected complexes. The binding in the NO complexes is similar to the
649 coordination of dioxygen to the metal centers, although the nitrosyl products are far more stable. There is, however,
650 no simple correlation between the rate and activation parameters and the selected donor groups or overall charge on
651 the Fe(II) complexes. It was found that high-stability of Fe(III) chelates do not form satisfactory redox catalysts and
652 that ligands that form suitable iron complexes as catalysts have both oxygen and nitrogen donors [25]. However, not
653 only the metal is oxidized, but also the ligands can be attacked [24]. To act as a catalyst in amine degradation, the
654 difference in stabilities of the Fe(III) and Fe(II) chelates must be low enough so that the Fe(III) form can be reduced
655 and that the Fe(III) chelate can be regenerated by the dissolved oxygen in the solvent at alkaline pH. The impact of
656 the ligands becomes clear when exemplarily the redox potential of [Fe(III)(edta)]⁻/[Fe(II)(edta)]²⁻ of 0.129 V is
657 compared with that of [Fe(III)(H₂O)₆]³⁺/[Fe(II)(H₂O)₆]²⁺ of 0.771 V. For the reduction of [Fe(III)(edta)]⁻, quite
658 strong reductants are needed. Chelating ligands are existing in aged CESAR1, like ethylenediamine (EDA) and oxalic
659 acid in sufficient amounts for the complexation of iron. Moreover, also complexes with positive charges are known
660 with less voluminous ligands, like the basic iron acetate [Fe(III)₃O(acetate)₆(H₂O)₃]⁺. It is unknown if the chelate
661 complexes given above or similar compounds are existing in the solvent. However, there is no doubt that the
662 potentially as catalyst acting metals in the solvent are complexed and that their accessibility for reaction partners and
663 their redox behavior are tuned by the ligands of the solvent matrix. A lower concentration limit for catalytically

active metal ions from which the catalytic effect can be expected to be negligible is not known. The solvent and the specific matrix of contaminants are more decisive for the degradation than the metal ion concentrations. Additionally, it is extremely unlikely that industrial-scale application of amine-based CO₂ capture could be realized with metal-free solvents. Even if capture plants could be completely built from non-metallic materials, as investigated in the project LAUNCH [7], already the pure technical amines will contain metal traces from their production processes and small metal or rust particles will enter the capture plant via the feed gas, so that metal ions will always be part of the solvent inventory. When cleaning of the solvent removes contaminants that hinder the effectivity of the catalysis, more damage than good is done. Excessive cleaning of the solvent would mainly waste valuable solvent.

4.3. Role of NO₂ for amine oxidation

If the removal of NO₂ from flue gas by pre-treatment based on scrubbing with a thiosulfate/sulfite solution can be realized in industrial scale and if this approach is able to significantly reduce the degradation of the solvent, depends on its physical and chemical properties. A very first indication for the ranking of the importance of NO₂ for solvent degradation compared to other oxidizing species in the gas phase can be derived from their molar concentrations in the flue gas. Values of Henry's law constants for aqueous solutions depend on the composition of the solution, i.e., on its ionic strength and on dissolved organics. In general, the solubility of a gas decreases with increasing salinity ("salting out"). An increase in ionic strength, mainly achieved by loading of the amines with CO₂ accompanied by protonation of the amines and to a lower degree by accumulation of ionic degradation products (e.g. organic acids R-COO⁻ residues and ammonium NH₄⁺) and trace compounds from the flue gas (e.g. dissolved Cl⁻, SO₂²⁻) leads to decreasing solubility of the gaseous species [28]. The O₂ solubility in amine-based solvents is typically in the range between 0.1 and 0.4 mmol/l for CO₂-unloaded solutions. At plant conditions with CO₂-loaded solutions at 40°C and above, the oxygen content is below 0.1 mmol/l and cannot be properly quantified up to now [29]. The low partial pressure of the nitrogen oxides and sulfur trioxide compared with molecular oxygen would suggest a special importance of oxygen for solvent degradation. Table 5 shows the typical molar concentrations of oxidizing species in the flue gas and some important properties for the mass transfer into the solvent and the chemical reactivity. For the nitrogen oxides the description of their oxidative reactivity after the dissolution in the

solvent is complicated [18]. NO and NO₂ undergo rapid interconversion in both gas and aqueous phases and can form N₂O₃ and N₂O₄. With increasing temperature, N₂O₄ is split into NO₂ (at 21°C 20%, 50°C 40%, 100°C 90%, 140°C 100% NO₂). The oxidation of NO is third-order overall, first-order in O₂ and second-order in NO. However, it is not a trimolecular reaction [32]. The oxidation of NO is third-order overall, first-order in O₂ and second-order in NO. However, it is not a trimolecular reaction [33]. The reaction path comprises two serial steps, first the formation of NO₃ that reacts in the next step with NO to N₂O₄. In the aqueous-phase, interconversion equilibria are established rapidly in comparison to the rates of the subsequent hydrolysis reactions with formation of nitrite and nitrate ions in the alkaline environment of the solvent, shown in Table 6.

Table 5. Concentrations of oxidative gaseous species in the flue gas at the entrance of the CO₂ absorber at the capture pilot plant at Niederaussem and Henry's Law Coefficients and Heat of Dissolution ΔH_{sol} at 298 K [31-34].

Species	Typical concentrations in flue gas [mol/m ³]	Henry's Law Constant in water [mol/(m ³ Pa)]	ΔH_{sol} [kJ mol ⁻¹]	Spin multiplicity/ unpaired electrons for ground state
O ₂	3 – 15	1.3 x 10 ⁻⁵	-12.1	3/2
NO	(3 – 57) x 10 ⁻³	1.9 x 10 ⁻⁵	-12.1	2/1
NO ₂	(0.04 - 0.16) x 10 ⁻³	1.2 x 10 ⁻⁴	20.9	2/1
SO ₃	<3 x 10 ⁻⁷	Fast, irreversible hydrolysis is assumed, which is equivalent to an infinite effective Henry's law constant	-	1/0

Table 6. Kinetics of some reactions of NO, NO₂, N₂O₄ solved in water at 293,15 K [18, 34-35].

Reaction	Rate constant k [mol ⁻¹ dm ³ s ⁻¹]
2NO _{2(aq)} ⇌ N ₂ O _{4(aq)}	(4.5 ± 1.0) x 10 ⁸ for initial [NO ₂] ranging from 8 to 30 x 10 ⁻⁶ mol ⁻¹ dm ³
NO _(aq) + NO _{2(aq)} ⇌ N ₂ O _{3(aq)}	1.1 x 10 ⁹ with excess [NO] ranging from 2 x 10 ⁻⁵ to 1 x 10 ⁻⁴ mol ⁻¹ dm ³
2NO _{2(aq)} /N ₂ O _{4(aq)} + H ₂ O ⇌ 2H ⁺ + NO ₂ ⁻ + NO ₃ ⁻	(6.5 ± 0.7) x 10 ⁷ for [NO _{2(aq)} /N ₂ O _{4(aq)}] from 0.46 to 6 x 10 ⁻⁶ mol ⁻¹ dm ³
(NO _(aq) + NO _{2(aq)})/N ₂ O _{3(aq)} + H ₂ O ⇌ 2H ⁺ + 2NO ₂ ⁻	7.4 x 10 ⁶

Consequently, it is not possible by studies of the kinetics of the latter reactions to ascertain whether the immediate precursor of reaction is a single molecule of the compound oxide or two molecules of the simple oxides. Under laboratory conditions, often high partial pressures of the nitrogen oxides are used for experiments to

709 accelerate degradation reactions and to achieve strong effects [36]. However, this means that second- and higher-
710 order reactions will be favoured and might mask any first-order reaction that runs in parallel, whereas under the real
711 conditions in the flue gas such a first-order mechanism might be dominant. Therefore long-time testing with
712 concentrations representative for real flue gas is important to evaluate the effect of NO₂ removal from the flue gas
713 on the solvent degradation.

714 Explanations of the results of the NO₂ removal campaign regarding the observed trend in the acid number
715 increase might be the negligible role of NO₂ as oxidant compared to oxygen, side reactions in the solvent that
716 consume NO₂ (e.g. dimerization to N₂O₄ followed by disproportion to NO₂⁻ and NO₃⁻ in the alkaline solvent) or
717 alternatively a too fast backformation of NO₂ from NO in the CO₂ absorber. It was observed that some “fresh” NO₂
718 is formed from NO - which is available in excess in the flue gas - after the pre-treatment on the way to the CO₂
719 absorber [10]. The processes in the CO₂ absorber (e.g. side reactions that consume NO₂, residence and life time of
720 NO₂) remain unclear. This knowledge gap could be filled by additional investigations on the NO/NO₂ equilibrium in
721 simulated flue gas (also containing representative concentrations of SO₂ and H₂O) with regard to the residence time
722 of the pre-treated flue gas (downstream NO₂-prescrubber, ducts, absorber) and a better understanding of the
723 reactions of NO₂ in an aged solvent [6]. The amount of NO₂ that might be formed inside the absorber remains
724 unclear and further investigations in the laboratory which go hand in hand with simulations of the NO₂ reduction
725 and formation along the flue gas path are needed. At the moment, the removal of NO₂ by scrubbing of the flue gas
726 with an alkaline thiosulfate solution seems to be ineffective.

727 In good agreement with investigations of the kinetics of nitrosamine decomposition, the concentrations of MNPZ
728 and to some degree of the total nitrosamines can be reduced by slightly elevated desorber temperature of 130°C.
729 Therefore, thermal treatment is a straightforward measure for nitrosamine control for the solvent CESAR1, based on
730 the high stability of the amines PZ and AMP and the low concentrations of NDELA which has a higher activation
731 energy for the degradation reaction than MNPZ [37]. For more thermal sensitive amines, like MEA, with the
732 tendency to form more NDELA and to suffer from increased oxidative degradation due to the higher temperatures,
733 other measures might be applicable. In aqueous solution, the rate of nitrosation of amines drops rapidly at pH higher
734 than pK_a of HNO₂ because of decreasing concentration of the active nitrosating species. No nitrosation caused by
735 aqueous nitrite has been observed at higher pH than 7.5. The pH of the lean CESAR1 solvent is 11.2 and of the rich

CO₂-loaded CESAR1 9.5. However, in [38] an alternative mechanism was investigated. Thus, MNPZ is probably formed through the nucleophilic attack of the nitrite anion on an iminium ion formed by reaction of PZ and aldehyde. Formaldehyde acts as catalyst. It was also proposed that no NO_x is necessary for the formation of nitrite as it is expected to be formed due to oxidation of NH₃ in the presence of •OH/O₂. Alternatively, also CO₂-catalyzed nitrosation by aqueous nitrite is possible.

Against this background the assumption that solvents using secondary amines would readily form stable nitrosamines in the capture plant, especially if no high level of reclaiming is used to remove degradation products and that the formation of nitrosamines is also simply linked with the concentration of NO_x in the flue gas must be re-evaluated. There is evidence from other pilot plant campaigns that support this finding [37].

5. Conclusion

The results from the 40 months campaign with CESAR1 at the capture plant in Niederaussem show that some suggested measures for solvent management need to be revised and cannot be generalized. It was demonstrated that the solvent matrix of the specific solvent and the accumulated contaminants which depend on the specific site conditions (flue gas composition, capture plant process configuration, plant materials, and operating characteristic) must be taken into consideration to better predict the degradation rate. Neither the solvent color nor the metal concentration in the solvent are appropriate measures to control solvent management technologies or to describe the performance of the solvent. Ion exchange can effectively remove anions and cations from the solvent, but the consequential effects on the solvent degradation rate must be carefully considered. Based on the results of the campaign with CESAR1 the removal of ligands from the solvent which form complexes with metal cations can increase the degradation rate. More investigation is needed regarding the interaction of metal cations, which have the potential to act as a catalyst for oxygen activation and solvent degradation, and degradation products with capability to form complexes to avoid the opposite of the expected effect when activating solvent management technologies. Also NO₂ removal from the flue gas upstream the CO₂ absorber by flue gas pretreatment with thiosulfate solution did not show a reducing effect on amine degradation in CESAR1.

The trade-off between a simplified laboratory experiment, to focus on postulated core processes of solvent degradation and the necessity to cover all relevant aspects of the real process configuration, comprises on one hand

the risk of oversimplification and on the other hand the challenge to understand and correctly connect all necessary details. For that reason, it is essential to evaluate results from laboratory tests and artificial short-time test campaigns on solvent degradation based on the facts received from long-time testing with real flue gases. Without proof of the efficacy of specific solvent management technologies for individual solvents in the real application case, theories on effect mechanisms and derived guiding principles for solvent management in capture plant operation can be misdirected. The proposed guidelines should be reviewed. This can contribute to avoid waste of valuable solvent and resources and to optimize the environmental benefit from amine-based carbon capture. An intelligent combination of complementary laboratory experiments - making also use of solvent from real operation - with meaningful pilot plant tests of sufficient duration and innovative approaches, like the head-to-head operation of CO₂ capture plants of different size realized in the project LAUNCH, offer the opportunity to clarify the effects caused by the interventions of solvent management and effective key factors for solvent degradation in industrial CO₂ capture.

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